

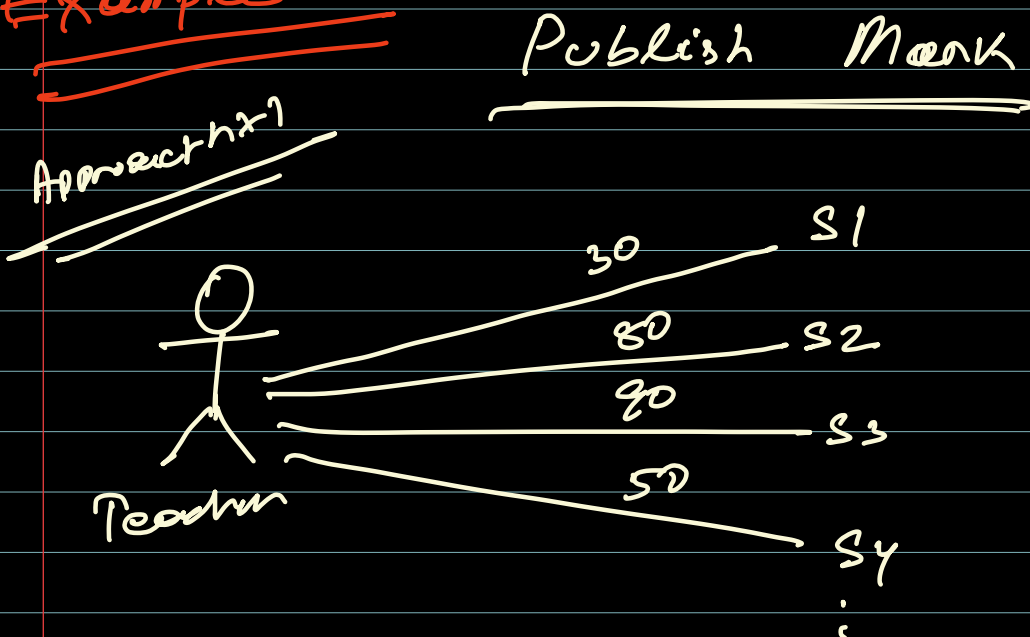
OBSERVER DESIGN PATTERN

- ① Definition
- ② Examples
- ③ Practical

① Definition

- ① Behavioural design pattern
- ② When the state of one object is changed all of its dependent objects are informed and updated
- ③ One to many

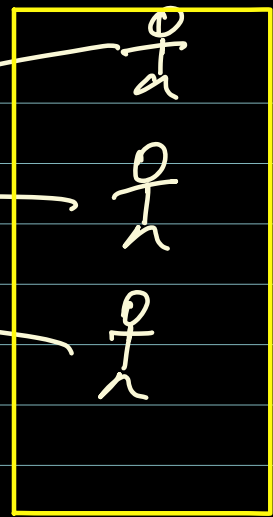
Example-1



Approach 2

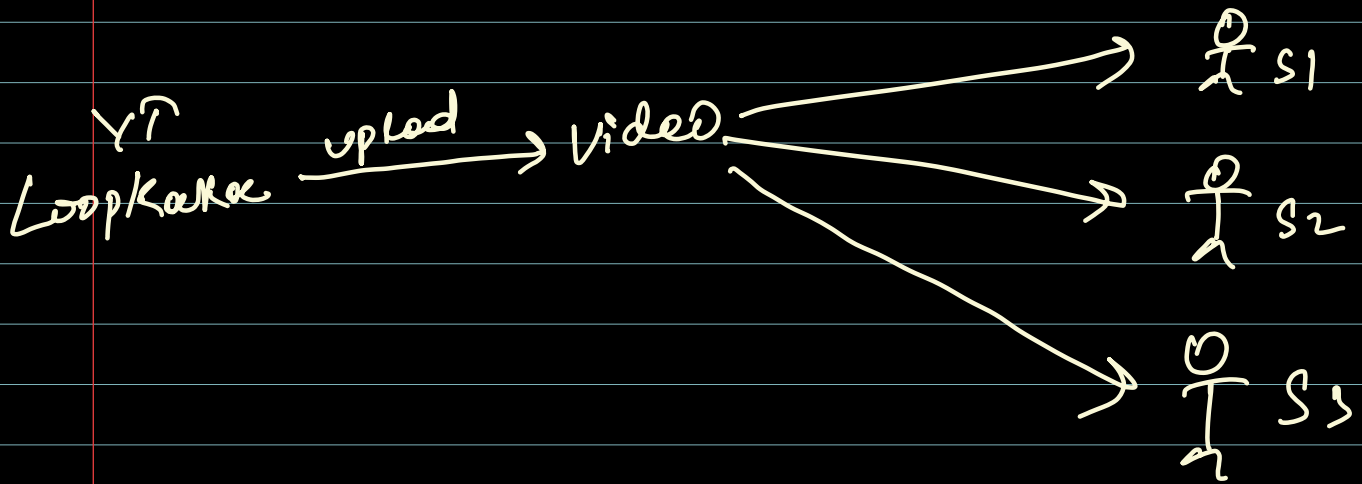


Notice	
S1 -	90
S2 -	30
S3 -	50
S4 -	70
:	:



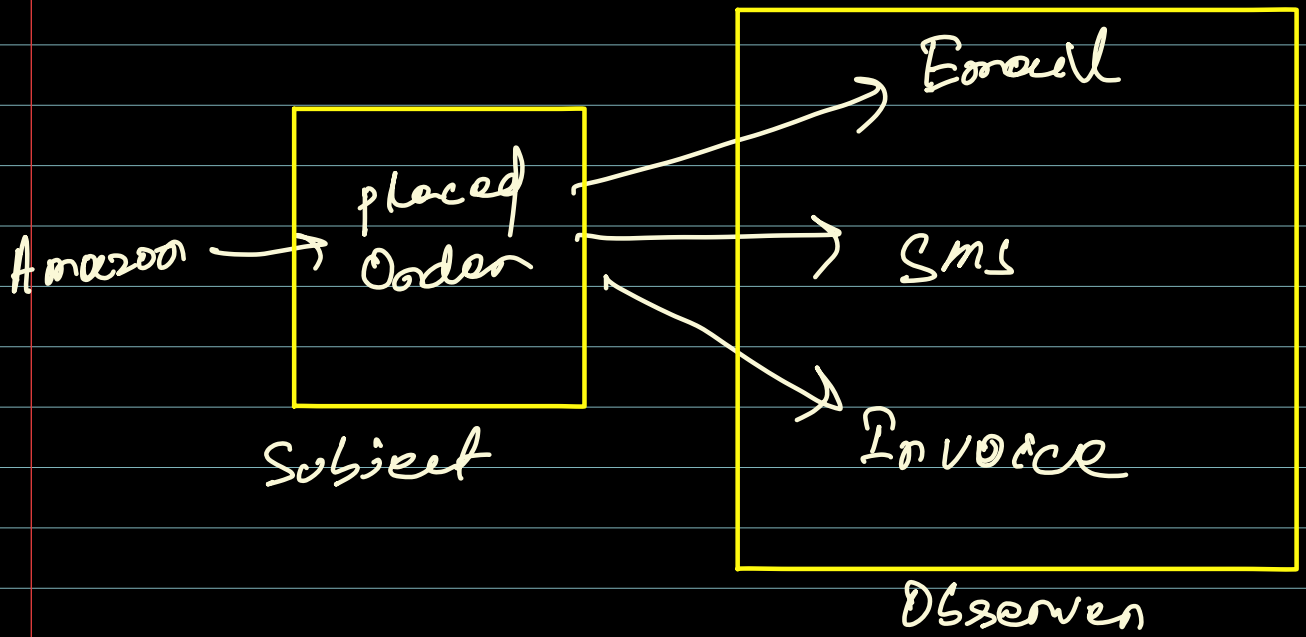
Example - 2

Youtube Video Upload



Example - 3

Amazon Order



```
public class EmailService {  
    public void sendEmail() {  
        System.out.println(x: "Send Email");  
    }  
}
```

```
public class InvoiceService {  
    public void generateInvoice() {  
        System.out.println(x: "Generate Invoice");  
    }  
}
```

```
public class SMSService {  
    public void sendSMS() {  
        System.out.println(x: "Send SMS");  
    }  
}
```

```
public class Amazon {  
    private EmailService emailService;  
    private SMSService smsService;  
    private InvoiceService invoiceService;  
  
    public Amazon(EmailService emailService, InvoiceService invoiceService, SMSService smsService) {  
        this.emailService = emailService;  
        this.invoiceService = invoiceService;  
        this.smsService = smsService;  
    }  
  
    public void orderPlace() {  
        emailService.sendEmail();  
        invoiceService.generateInvoice();  
        smsService.sendSMS();  
    }  
}
```

Problems

① DIP

② OCP

③ code change & re-deploy

